Mak Chun Chi

Analyst Programmer · 6+ years experience

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ISC2 CISSP Study Notes & Q&A (Accreditation to *Deepseek* & *Manus*)

CompTIA Network+ (N10-009) Study Notes (Accreditation to *Deepseek* & *Manus*)

CORE COMPETENCIES

Languages

C# · Java · JavaScript · HTML5/CSS3 · PHP

Frameworks

.NET · Spring Boot · React · Node.js · Express

Databases

MySQL · SQL Server

DevOps & Tools

Docker · Git · REST APIs · Socket Programming · WordPress

Expertise

Root Cause Analysis · Technical Documentation · Input Validation · Legacy Modernization

EDUCATION

Bachelor of Engineering in Electrical Engineering

The Hong Kong Polytechnic University · 2014 – 2016

Higher Diploma in Electrical Engineering

The Hong Kong Polytechnic University · 2012 – 2014

CERTIFICATIONS

CompTIA Network+ (N10-009) – 2026

CERTIFICATE OF COMPLETION

ISC2 CISSP (Udemy) – 2026

CompTIA Networking+ (Udemy) – 2026

ACADEMIC PROJECTS

Solar Energy Optimization System (Bachelor Thesis)

Optimised solar panel efficiency through load analysis and tracking algorithms (PolyU 2015-2016).

Embedded Line-follower Robot (Higher Diploma)

Developed a driverless car using sensor fusion and microcontroller logic (PolyU 2013-2014).

Analyst Programmer with **6+ years of full-stack development** and system maintenance expertise. Specialised in **C# .NET, Java Spring Boot, LAMP, React/Node.js**. For the past four years, I have built and maintained **earthquake information systems for the Hong Kong Observatory**, driving Docker containerisation, technical documentation, and root cause analysis to enhance platform reliability.

EMPLOYMENT & PROJECT EXPERIENCE

Analyst Programmer @ IT Solutions Ltd. (Hong Kong Observatory)

Jul 2025 – Present

- Developed a **React + Express + Node.js** web application to collect and visualise locally felt earthquake reports, deployed with **Docker** inside a secured internal network.
- Authored technical documentation and user guides; performed proactive root cause analysis and preventive fixes to reduce recurring incidents.

Analyst Programmer @ Kelly Services (Hong Kong Observatory)

Jun 2022 – Jun 2025

- Built a full-stack earthquake data dissemination platform using **PHP/LAMP** stack running inside Docker; enforced client-side validation.
- Led **root cause analysis (RCA)** initiatives and implemented preventive patches, significantly reducing recurrence of system failures.
- Managed WordPress-based technical documentation hub; standardised user guides for 3 internal applications, adopted as team-wide reference.

Solution Developer @ PCCW

Sep 2019 – Jun 2022

- Developed a web-based **booking appointment system** using **C# .NET**, supporting real-time availability validation, rescheduling, and cancellations.
- Built a **Windows Forms** desktop application as a query interface for backend databases.
- Implemented a **multithreaded socket server in Java** for concurrent client connections in real-time data exchange.
- Collaborated on a **Spring Boot workflow management system**.

KEY ACHIEVEMENTS

Containerisation Migration ⚡ IMPACT: DEPLOYMENT CONSISTENCY

Migrated two legacy earthquake reporting systems to **Docker + LAMP / MERN stacks**, improving deployment consistency, rollback capability and maintainability across HKO infrastructure.

Reliability Improvement 📉 REDUCING INCIDENT RECURRENCE

Reduced recurring system incidents through proactive monitoring, root cause analysis (RCA) and preventive fixes, directly enhancing HKO platform stability and user trust.

Documentation Standardisation 📘 TEAM-WIDE ADOPTION

Authored **comprehensive user guides and technical runbooks** for 3+ internal applications, adopted as the official team standard.

Cybersecurity Upskilling 🔐 SECURITY-FIRST MINDSET

Completed **CompTIA Network+** and **ISC2 CISSP** core curriculum.